ELSEVIER

Cell Calcium 37 (2005) 425–432

cell calcium

www.elsevier.com/locate/ceca

Retinal waves: mechanisms and function in visual system development

Sally I. Firth, Chih-Tien Wang, Marla B. Feller *

Neurobiology Section, Division of Biological Sciences, Pacific Hall 3127, University of California at San Diego, 9500 Gilman Dr, La Jolla, CA, 92093-0357, USA

Received 18 December 2004; accepted 6 January 2005

Abstract

A characteristic feature of developing neural networks is spontaneous periodic activity. In the developing retina, retinal ganglion cells fire bursts of action potentials that drive large increases in intracellular calcium concentration with a periodicity of minutes. These periodic bursts of action potentials propagate across the developing inner retina as waves, driving neighboring retinal ganglion cells to fire in a correlated fashion. Here we will review recent progress in elucidating the mechanisms in mammals underlying retinal wave propagation and those regulating the periodicity with which these retinal waves occur. In addition, we will review recent experiments indicating that retinal waves are critical for refining retinal projections to their primary targets in the central visual system and may be involved in driving developmental processes within the retina itself.
© 2005 Elsevier Ltd. All rights reserved.

Keywords: Calcium transients; Acetylcholine; Retina; Development; Periodic spontaneous activity; Patterned activity

1. Introduction

Periodic spontaneous activity is found in many parts of the developing central nervous system including the spinal cord, cortex, hippocampus and retina (reviewed in [1]). There is growing evidence that these early spontaneous depolarizations and increases in intracellular calcium play a crucial role in driving many aspects of development, including axon guidance, establishment of cell phenotype, local circuit formation and establishment of sensory maps. This phenomenon has been studied extensively in the retina. During a developmental period prior to vision, when there is tremendous sculpting of circuits within the visual system, immature retinal circuits spontaneously generate propagating bursts of action potentials termed retinal waves. Manipulations of retinal waves affect several features of early visual circuits, indicating that spontaneous retinal activity plays a critical role in driving activity-dependent processes during visual system development [2,3].

It has been more than two decades since the phenomenon of neuronal retinal waves was first described in the developing mammalian retina (for complete references see [4]). Spontaneous bursts of action potentials that occur roughly once per minute were recorded in the neonatal rabbit retina and were blocked by nicotinic acetylcholine receptor (nAChR) antagonists indicating a role for early cholinergic circuits in driving the activity. Over 10 years later, in vivo recordings in fetal rat pups revealed that neighboring retinal ganglion cells (RGCs) fire correlated bursts of action potentials. Multielectrode array experiments showed these periodic correlated bursts of action potentials propagate like waves across the ganglion cell layer of ferret retinas. Calcium imaging demonstrated correlated calcium transients also propagate across the ganglion cell layer of the retina in a wave-like manner. Simultaneous electrophysiological recordings and calcium imaging experiments indicate that these calcium transients reflect the calcium influx during depolarization and not release from intracellular stores. Therefore, retinal waves recorded by calcium imaging correspond to waves of depolarization.

Calcium imaging enables the pattern of spontaneous activity to be analyzed over much larger regions of the retina than recorded with the multielectrode recordings. This has allowed investigators to define several spatiotemporal properties of waves (Fig. 1, [5–7]). Waves initiate at random locations and propagate over a finite distance of the retina. The

* Corresponding author. Tel.: +1 858 822 4273; fax: +1 858 534 7309.
E-mail address: mfeller@ucsd.edu (M.B. Feller).

0143-4160/$ – see front matter © 2005 Elsevier Ltd. All rights reserved.
doi:10.1016/j.ceca.2005.01.010

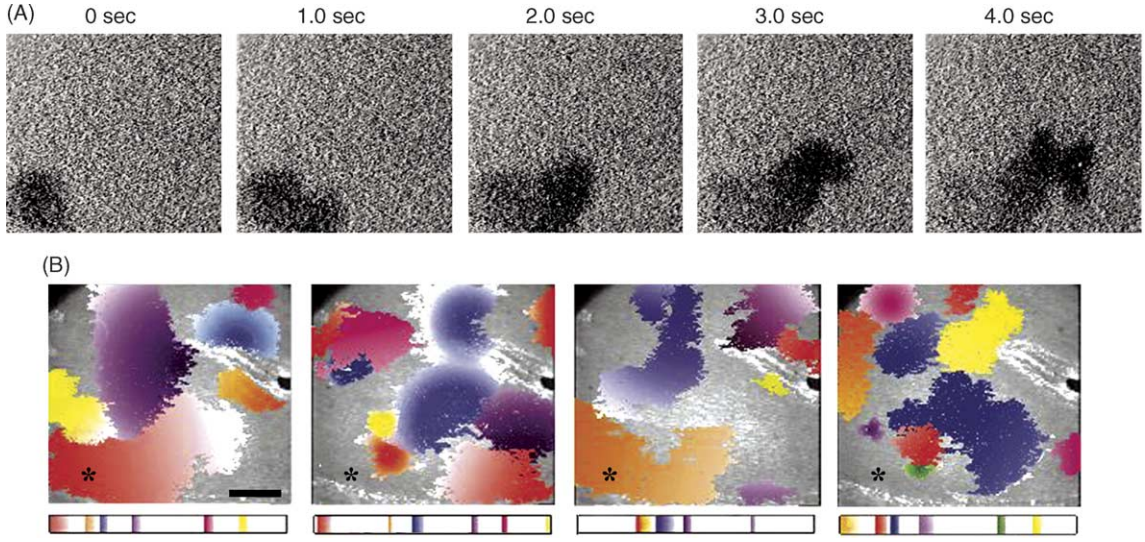


Fig. 1. Spatiotemporal properties of spontaneous retinal waves. (A) Successive 1-s intervals from a movie of fluorescence imaging from a P2 ferret retina labeled with the calcium indicator fura-2AM, where fluorescence decreases are associated with increased $[Ca^{2+}]_i$. Retinal waves propagate in a saltatory fashion and travel with average velocities ranging from 100–300 μm/s. (B) Summary of four consecutive minutes of activity. Background: fluorescence image of retina incubated in fura-2AM. Different colors correspond to different waves with the color-coded time bar below each frame indicating the time of initiation of each wave. The gradation in color roughly corresponds to the direction of propagation (dark to light). The red domain is the wave shown in part (A) Wave boundries do not repeat and are determined in part by a refractory period. A local region (✻) of the retina participates roughly once per minute (red wave in the first frame and orange wave in the third frame). Scale bar = 200 μm (adapted from [5]).

domain, defined by the propagation boundaries for individual waves, is determined in part by a refractory period—a period of time lasting 40–60 s following a wave during which time subsequent retinal waves cannot propagate within the same area of the retina.

A biophysical model of the developing retina predicts that this refractory period dictates several features of retinal waves, including the velocity and the periodicity of retinal waves [6,8]. Velocity is defined as the distance between wave fronts per unit time, and the periodicity is the average time interval between wave-induced calcium transients or depolarizations recorded in a particular neuron in the ganglion cell layer. According to this model, wave velocity is determined by the density of refractory cells—if there is a low density of refractory cells, waves propagate quickly and if there is a high density, waves propagate more slowly. As the density of refractory cells goes above a threshold, waves are prevented from propagating at all, and therefore a high density of refractory cells defines the boundaries of waves. This also means that shortening the length of the refractory period will increase the velocity of waves. The relationship between the refractory period and wave periodicity is more straightforward—a given retinal neuron cannot participate in sequential waves at intervals shorter than the refractory period. Hence, a major prediction of the model is that the same manipulations that increase wave velocity should also decrease the interval between retinal waves.

Here, we review current insights into the mechanisms underlying retinal wave generation and a role for retinal waves in driving refinement of circuits involving RGCs. (Note, we focus primarily on mammalian retinal waves, which may have distinctly different mechanisms and functions from the non-mammalian retina.) Although progress has been made in defining the cell types that are involved in generating retinal waves [2–4], several general questions remain regarding the cellular basis of this patterned activity. We will explore the mechanisms that determine the velocity and periodicity of retinal waves because these properties dictate the spatiotemporal properties of the correlated activity that drive downstream developmental processes. Velocity determines the temporal structure of correlations between the neighboring RGCs that may be critical for retinotopic refinement. The periodicity of action potential-driven calcium transients may be critical for the “frequency tuning” of calcium transients that drive a variety of early developmental events (see Spitzer in this issue). We also review the experiments that address whether these correlated firing patterns associated with retinal waves are critical for refinement of retinal projections to the dorsal lateral geniculate nucleus (dLGN) of the thalmus and the superior colliculus (SC) as well as the development of the retina itself.

2. Cellular basis of retinal wave propagation and periodicity

What determines the velocity of retinal waves? Retinal waves propagate in a saltatory manner, with an average velocity of 100–300 μm/s [4]. This ‘intermediate’ speed of propagating neural activity is rarely seen in the nervous system. This velocity by far exceeds the rate of extracellular diffusion of excitatory substances, such as those underlying spread-

ing depression, a well-characterized phenomenon in the adult retina [9]. In addition, the observed propagation velocity is significantly slower than epileptiform waves that are induced by GABA-A receptor antagonists in the hippocampus or cortex [10]. In this section, we review the synaptic circuitry that is known to be involved in retinal wave generation and the function these circuits may have in regulating the propagation velocity.

Retinal waves are mediated by several mechanisms [2–4]. The neurotransmitter receptors required for retinal wave generation switch from neuronal nAChRs ('early stage') to ionotropic glutamate receptors ('late stage'). In addition to the fast excitatory neurotransmitter ACh, the neuromodulator adenosine, functioning through a cAMP/PKA second messenger cascade is involved in retinal wave propagation [11]. Last, gap junction coupling in the inner retina is likely to be involved in retinal wave propagation since gap junction blockers at high concentrations block waves and at lower concentrations decrease the domain size [12]. A detailed summary of these synaptic circuits is schematized in Fig. 2.

To determine the relative contribution of synaptic transmission, gap junctions and adenosine release, a novel form of propagating activity observed in the presence of an L-type Ca^{2+} channel agonist (FPL64176) was studied [12,13]. In FPL, waves propagate with almost twice the velocity of control waves. Interestingly, FPL-waves persist in the presence of fast neurotransmitter antagonists, including nAChR, AMPA, NMDA, glycine and GABA-A receptor antagonists [12]. However, as with normal retinal waves, these FPL waves are inhibited by gap junction blockers and abolished by adenosine antagonists or adenosine deaminase. These results suggest several hypotheses. First, retinal wave velocity is influenced by the amount of calcium influx through L-type calcium channels. Second, wave propagation can occur in the absence of fast synaptic transmission. Third, the primary site of adenosine-mediated modulation of retinal waves is downstream of fast synaptic transmission and may

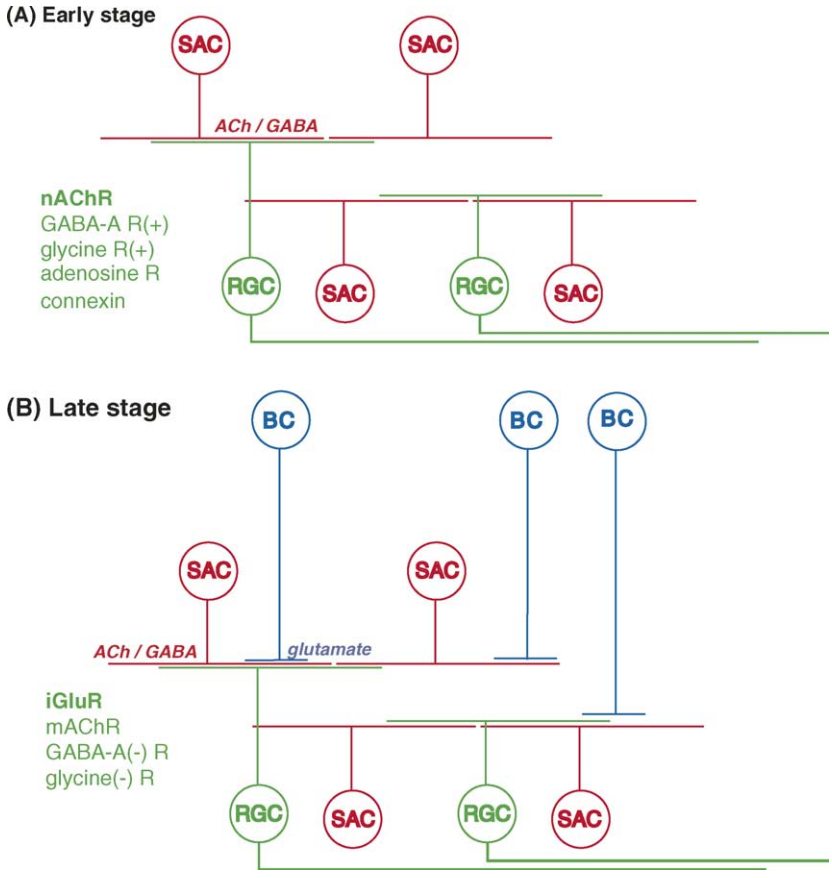


Fig. 2. Neural circuits that mediate mammalian retinal waves. (A) Schematic of the circuitry that mediates waves during an early stage of retinal waves, defined as P0–P14 in ferrets and P0–P11 in mice, E26–P0 in rabbits. Retinal ganglion cells (RGCs, green) receive synaptic inputs from starburst (cholinergic) amacrine cells (SAC, red). SAC also release GABA. Retinal waves are abolished by blockade of neuronal nAChRs [3,21]. On the left of the schematic is a list of the receptors that may modulate the activity of the color-coded cell type. Gap junctions (composed of connexins) and adenosine are also involved in retinal wave generation, though the exact circuitry for the contributions of gap junctions and adenosine are yet to be determined. (B) Schematic of the circuitry that mediates waves during the late stage of retinal waves, defined as P15–P28 in ferrets, P11–P21 in mice and P1–P9 in rabbits. Glutamatergic terminals from bipolar cells (BC, blue) first appear and begin to release glutamate in the IPL [2]. Ionotropic glutamate receptor antagonists, block retinal waves [22,36] and cholinergic neurotransmission switches from nicotinic to muscarinic receptor-driven activity [23]. A role for adenosine and gap junctions is yet to be elucidated.

be due to either modulation of the postsynaptic site or an increase in the strength of gap junction coupling. Whether the site of FPL activation is presynaptic to adenosine release remains to be determined.

We return to the question, what is the rate-limiting step that determines the retinal wave velocity of 100–300 μm/s? One possibility is that the efficacy of electrical synapses may be dynamically modulated by intracellular calcium or cyclic AMP (cAMP) concentrations. Second messenger modulation of coupling has been observed thoughout the retina [14], though no one has yet determined the time course over which this modulation occurs. A second possibility is that synaptic transmission may involve diffuse release of a neurotransmitter, such as ACh or adenosine. During the developmental period when retinal waves occur, ACh esterase activity is low and therefore ACh may be diffusing into the extracellular space and depolarizing cells at a distance. Consistent with this hypothesis, ACh released in the inner retina diffuses to the outer retina and stimulates muscarinic AChRs in the developing ventricular zone [15]. A third possibility is that the action adenosine may be the rate-limiting step. Activation of adenosine receptors, which are G-protein coupled may directly, or via a second messenger cascade, activate a depolarizing conductance [16]. Further experiments are required to distinguish these possibilities.

The biophysical model [6,8] predicts that the same manipulations that increase wave velocity should also increase wave frequency. This is indeed the case. Adenosine receptor agonists, adenylate cyclase activators, elevated levels of extracellular potassium and L-type calcium channel agonists, all of which increase wave velocity, increase wave frequency [6,11,12]. Adenosine deaminase, adenylate cyclase or PKA inhibitors, gap junction blockers, all of which decrease wave velocity, and decrease wave frequency [11,12]. These consistent affects on wave velocity and periodicity indicate that the seemingly complex effects different manipulations have on retinal waves may all be explained by alterations to the refractory period.

Several manipulations of wave periodicity exist, however the underlying pacemaker that dictates this periodicity is unknown. There are two general categories of mechanisms that could underlie this slow periodicity. The first category is cell-autonomous mechanisms where the oscillations are generated in an individual cell. The most commonly studied cell-autonomous pacemakers are oscillations in membrane potentials that are driven by time-varying voltage-gated conductances, such as those that underlie cardiac pacemakers. Since in the intact retina, RGCs participate in waves once a minute, this model would predict that cells within the retina would have minute-long periods of hyperpolarization. In this scenario, hyperpolarization would represent the refractory period. However, current clamp recordings from RGCs and starburst amacrine cells have not revealed spontaneous oscillations in membrane potential [4]. Another possibility is that cell autonomous pacemakers are not reflected by changes in membrane conductance but perhaps in intracellular signaling. For example, calcium oscillations in glial cells are mediated by spontaneous release of calcium from intracellular stores [17]. Though such slow oscillations in calcium have not been observed in retinal neurons, there may be oscillations in other second messengers, such as cAMP, that may mediate these responses.

A second category of pacemaker mechanisms is where the slow periodicity is an emergent property of network interactions. For example, periodic activity can emerge from reciprocally connected excitatory and inhibitory neurons, such as those found in the adult spinal cord. A network-based model composed of exclusively of excitatory synapses has also been proposed to explain the slow periodicity observed in developing neural circuits [18,19]. In these models, recurrent excitation via synaptic connections causes growth in overall level of activity in a large population of neurons. When the activity level surpasses a threshold, an event (e.g. a retinal wave) occurs. Network activity is then quenched by the slow negative feedback of network depression, such as synaptic depression, a term used to describe a transient reduction in strength of synaptic transmission. In this scenario, network depression would represent the refractory period.

There is some evidence that network properties modulate wave periodicity. Gamma aminobutyric acid (GABA) and glycine play a modulatory role in the generation of retinal waves (Fig. 2). During the early stages of retinal waves, GABA and glycine are excitatory and later in postnatal development they become inhibitory [20]. GABA-A receptor antagonists increase wave frequency during the late stage of retinal waves indicating GABA-A receptors are inhibitory in this developmental period [2,21]. Though GABA-A receptors are depolarizing in the early stage, their effects on retinal waves are unclear [11,22,23]. The modulation of retinal waves by glycine shifts from functionally excitatory to inhibitory during the switch from early to late retinal waves. Blockade of glycinergic receptors by strychnine abolishes retinal waves in the early stage, whilst strychnine application increases wave frequency in the late stage of retinal waves [21]. Hence, retinal interneurons may influence wave periodicity by modulating the excitability of the network.

Whether the pacemaker underlying wave periodicity is an emergent property of the immature network or a single pacemaker neuron is not yet determined. To better isolate cell autonomous or network-based pacemakers, we have developed a culture model. For cultures of dissociated retinal neurons grown under conditions where there is robust survival of RGCs [24], fast glutamatergic synaptic connections between RGCs drive rhythmic bursts of spontaneous synaptic activity. The periodicity of these compound synaptic events is 5–10 s, and matches the recovery from synaptic depression [25]. This network-based correlated activity has also been shown in dissociated cultures of glutamatergic neurons from other parts of the central nervous system [26,27]. However, intervals of 5 s are shorter than the periodicity in the intact developing retina. Recently, we demonstrated that dis-

sociated retinal cultures grown in the presence of media that contains cells from the superior colliculus display periodic spontaneous activity with several features similar to the intact retina [28]. By using this dissociated cell system, we hope to elucidate the cellular basis of the slow periodicity of retinal waves.

3. A role for retinal waves in visual system development

Many features of the retinal waves, as characterized by calcium imaging and extracellular multielectrode array recordings, suggest that they may drive early activity-dependent formation of functional circuits (reviewed in [2]). First, retinal waves are present during the period when the inner plexiform layer (IPL) is stratifying into the five layers seen in the adult retina and there is extensive refinement of retinal axons in their central visual targets, the dLGN and SC. Second, retinal waves propagate slowly across the retina with large wave fronts; therefore action potentials in nearby RGCs are more correlated than distant RGCs. Third, the duration of action potential bursts (a few seconds) are relatively short compared to the quiescent periods (minutes); hence, it is unlikely that RGC activity originating from different eyes would be temporally correlated. Based on the assumption that RGCs that fire together, wire together [29], retinal waves provide the ideal patterned activity to drive retinotopic or eye specific refinement of the RGC projections to central visual centers. However, whether the correlated firing patterns of retinal waves are essential for these processes is still debated.

3.1. Retinotopic refinement of retinocollicular axons

Retinotopic refinement of retinocollicular projections has been extensively studied (for recent review see [30]; Fig. 3). The bulk of the refinement in mammals occurs during the same developmental period as early stage retinal waves [31–34]. Postnatal refinement of RGC axons is thought to be activity-dependent because NMDA receptor antagonists in the SC disrupt normal retinocollicular refinement during this developmental period [35].

McLaughlin et al. [34] tested whether early-stage retinal waves drive activity-dependent refinement of retinocollicular projections using mice lacking the β2 subunit of the nAChR, a receptor involved in retinal wave generation [36]. β2−/− mice have no early stage waves, however multielectrode recordings reveal the persistence of uncorrelated action potentials [34]. In P8 β2−/− mice retinocollicular axon arborizations are significantly larger than in wildtype mice. This lack of retinotopic refinement of retinocollicular projections persisted after the period of glutamatergic driven retinal waves, which start precociously at P9 in β2−/− mice [34,36]. Together, these results suggest the cholinergic driven waves are required for axon refinement in the SC and that these early stage retinal waves represent a critical period for retinotopic refinement.

3.2. Eye-specific segregation of retinogeniculate projections

Retinogeniculate projections from opposing eyes are initially intermixed in the dLGN and become segregated during the developmental period when retinal waves are present [31]. Several experiments support the hypothesis that the segregation of retinogeniculate projections into eye-specific layers requires neural activity in the retina (Fig. 3). A continuous infusion of tetrodotoxin (TTX) above the optic chiasm to block transmission to the dLGN prevents the segregation of these retinogeniculate afferents in cats [37]. When retinal waves are blocked from P0 to P8 in ferret by repeated binocular intravitreal injections of the noncompetitive nAChR agonist, epibatidine, retinogeniculate projections no longer segregate into eye-specific layers in the dLGN [38,42,50]. Therefore, if the transmission of all spiking activity from the RGCs is prevented from reaching the dLGN, eye-specific layers no longer form in the dLGN. However, in contrast to the previous results, intraocular injections of TTX either monocularly or binocularly into ferret eyes, inhibit but do not completely prevent eye specific segregation of retinogeniculate afferents [39]. Since TTX blocks action potentials but not all calcium transients induced by retinal waves [11], one interpretation of these results is that periodic calcium transients and not action potentials are the relevant feature of retinal waves for driving eye-specific refinement (as discussed in [40]).

There is also evidence that eye specific segregation in the dLGN is due to activity-dependent competition between the RGCs projecting from opposing eyes. Monocular injections of epibatidine prevent eye-specific layers from forming and the area of the dLGN covered by the more active eye increases dramatically [38]. Stellwagen et al. [41] increased the frequency, velocity and domain size of retinal waves in one retina by elevating cAMP levels, and found that the dLGN territory covered by the more active retina is increased but segregation is normal. When activity is elevated in both eyes there is no change in extent of eye-specific segregation or size of the ipsilateral or contralateral domains in the dLGN [41].

These results support the idea that activity-dependent competition between retinal projections is important for axon refinement in the dLGN but also suggest that the precise pattern of firing induced by retinal waves is not critical for driving this process. This latter hypothesis is supported by a manipulation where Huberman et al. [40] use an immunotoxin to deplete cholinergic amacrine cells from neonatal ferret retina. This procedure interrupts the periodic spontaneous activity in the RGCs and uncorrelated spontaneous spiking activity remains. Under these conditions, segregation of retinogeniculate projections in the dLGN remain normal [40]. This result suggests that disrupted firing of action potentials, which leads to reductions in nearest neighbor correlations, are sufficient to drive eye specific segregation in the dLGN.

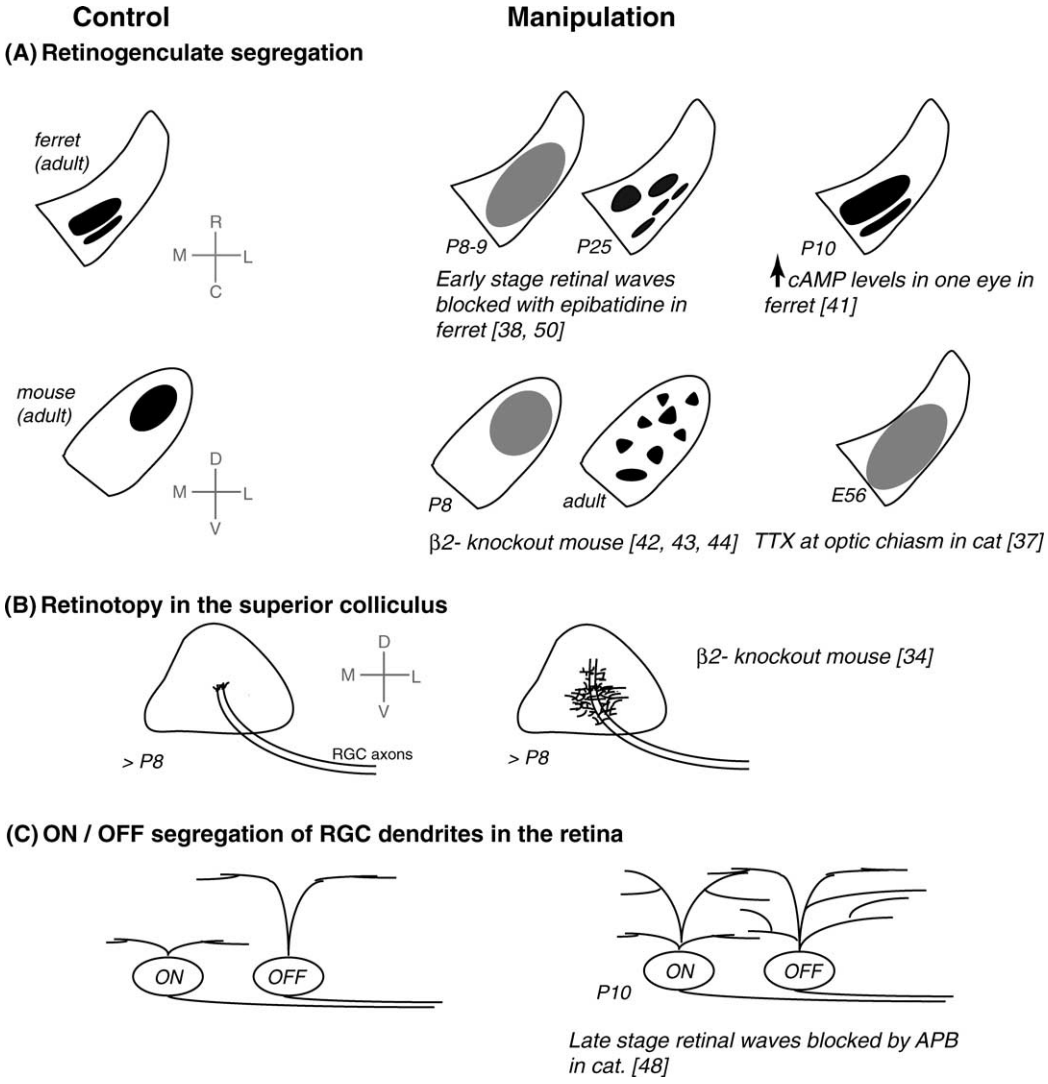


Fig. 3. Summary of several known functions of retinal waves. (A) Schematics of the normal anatomical patterning of retinogeniculate terminals in sections of the dorsal lateral geniculate, in control (left) and manipulated (right) conditions for a variety of mammalian species. Patterns of retinogeniculate projections are shown for ipsilateral dominant (black) and contralateral dominant (white) or mixed (grey) projections. D = dorsal, M = medial, L = lateral, V = ventral, R = rostral and C = caudal. (B) Schematic of retinotopic refinement of retinocollicular projections, in sections of the superior colliculus, in control (left) and β2−/− mice (right) that lack early stage retinal waves. (C) Schematic of RGCs in vertical section showing the ON and OFF sublamina of RGC dendrites, in control (left) and after P10 in cats in which had late stage (glutamatergic) waves were blocked (right).

In contrast to the above hypothesis, studies using β2−/− mice suggest that patterned activity in the retina is essential for normal eye-specific segregation of retinogeniculate axons. By the end of the first postnatal week, β2−/− retinogeniculate axons fail to segregate into eye-specific layers [42,43]. However, unlike the retinocollicular projections, late-stage glutamatergic-driven retinal waves can drive eye-specific segregation into an unlayered, patchy pattern [43,44]. This result indicates that activity-dependent plasticity can still occur in β2−/− mice, though the first postnatal week constitutes a critical period for segregation into the stereotypical layered pattern.

Is patterned activity required for eye-specific segregation? The resolution of this controversy may be that the altered spiking patterns induced by immunotoxin-depletion of cholinergic interneurons and adenylate cyclase activators still retain several key features of retinal waves. Indeed, in both cases, calcium imaging revealed that propagating waves still persisted during the treatments, in contrast to β2−/− mice where correlated calcium increases are completely prevented between P1 and P8 [36]. The identity of the key features of retinal waves that drive eye-specific refinement are yet to be determined.

3.3. A potential role for retinal waves in development of the local retinal circuits

Retinal waves may drive the ON and OFF sublaminae segregation of RGC dendrites [45]. In the IPL, RGC dendrites segregate into five strata. The two innermost strata comprise

the ON sublamina and the two outermost strata are the OFF sublamina, responding most robustly to increases and decreases in illumination, respectively. Though reaching adult levels of ON–OFF segregation of RGC dendrites requires visual experience [46], a significant amount of segregation can be observed before vision occurs [36,47].

Prior to vision, intraocular injections of aminophosphonobutyric acid (APB) prevent normal ON- and OFF-segregation of dendrites from β class of RGCs in cats [48]. Though in adult animals, APB blocks activity in the ON pathway, prior to eye-opening APB significantly inhibits all spontaneous calcium transients in the developing retina [49]. The APB injections were given between P2 and P8 in cat, which corresponds to late-stage retinal waves. Therefore, these results suggest that glutamate-driven retinal waves may be essential to drive ON and OFF segregation of RGC dendrites prior to eye opening.

Bansal et al. [36] examined the involvement of the early-stage retinal waves in the ON and OFF segregation using the β2−/− mouse. The stratification of starburst amacrine cell dendrites is normal in these mice. In addition, RGC stratification into ON–OFF layers still occurs, however the formation of distinct strata is delayed. After the developmental switch to glutamatergic-receptor driven retinal waves, the RGC dendrites stratified normally [36]. Hence, early stage waves are likely to play a more limited role in RGC dendritic stratification, perhaps in particular classes of RGCs that stratify earlier in development [47].

There is also intriguing new evidence that retinal waves may be critical for more basic cellular developmental processes. For example, elevated levels of cAMP driven by retinal activity are critical for RGC survival [24]. In addition, retinal waves drive retrograde signaling to the ventricular zone during the early development of the retina, inducing dramatic correlated increases in intracellular calcium in retinal progenitor cells [15]. Hence retinal waves may also play a crucial role in coordinating retinal cytogenesis.

In this review, we provide an overview of the recent results examining the cellular mechanisms underlying retinal wave propagation and periodicity – two features critical for determining the correlations induced in spontaneous activity in developing RGCs –and the evidence that these correlation patterns are critical for the normal refinement of retinal projections to the dLGN and SC. A role for retinal waves in the development of other primary retinal targets and the development of the retina itself requires further investigation.

Acknowledgements

Supported in part by the Whitehall Foundation, March of Dimes, McKnight Scholars Fund, and the National Institute of Health (grant no. NS13528-01A1). Thanks to C.L. Torborg, T. del Rio and K.A. Hansen for their critical comments on the manuscript.

References

[1] M.B. Feller, Spontaneous correlated activity in developing neural circuits, Neuron 22 (1999) 653–656.
[2] R.O.L. Wong, Retinal waves and visual system development, Annu. Rev. Neurosci. 22 (1999) 29–47.
[3] M.B. Feller, The role of nAChR-mediated spontaneous retinal activity in visual system development, J. Neurobiol. 53 (2002) 556–567.
[4] Z.J. Zhou, The function of the cholinergic system in the developing mammalian retina, Prog. Brain Res. 131 (2001) 599–613.
[5] M.B. Feller, D.P. Wellis, D. Stellawagen, F.S. Werblin, C.J. Shatz, Requirement for cholinergic synaptic transmission in the propagation of spontaneous retinal waves, Science 272 (1996) 1181–1197.
[6] M.B. Feller, D.A. Butts, H.L. Aaron, D.S. Rokhsar, C.J. Shatz, Dynamic process shape spatiotemporal properties of retinal waves, Neuron 19 (1997) 293–306.
[7] E. Sernagor, S.J. Eglen, M.J. O'Donovan, Differential effects of acetylcholine and glutamate blockade on the spatiotemporal dynamics of retinal waves, J. Neurosci. 20 (2000) RC56.
[8] D. Butts, M.B. Feller, C.J. Shatz, D. Rokhsar, Retinal waves are governed by collective network properties, J. Neurosci. 19 (1999) 3580–3593.
[9] H. Martins-Ferreira, M. Nedergaard, C. Nicholson, Perspectives on spreading depression, Brain Res. Rev. 32 (2000) 215–234.
[10] D.A. McCormick, D. Contreras, On the cellular and network bases of epileptic seizures, Annu. Rev. Physiol. 63 (2001) 815–846.
[11] D. Stellwagen, C.J. Shatz, M.B. Feller, Dynamics of retinal waves are controlled by cyclic AMP, Neuron 24 (1999) 673–685.
[12] J.H. Singer, R.R. Mirotznik, M.B. Feller, Potentiation of L-type calcium channels reveals nonsynaptic mechanisms that correlate spontaneous activity in the developing mammalian retina, J. Neurosci. 21 (2001) 8514–8522.
[13] C.L. Torborg, C.-T. Wang, G. Muir-Robinson, M.B. Feller, L-type calcium channel agonist induces correlated depolarizations in mice lacking the b2 subunit nACHRs, Vision Res. 44 (2004) 3347–3355.
[14] D. Becker, V. Bonness, P. Mobbs, Cell coupling in the retina: patterns and purpose, Cell Biol. Int. 22 (1998) 781–792.
[15] M.M. Syed, S. Lee, S. He, Z. Zhou, Spontaneous waves in the ventricular zone of developing mammalian retina, J. Neurophysiol. 91 (2004) 1999–2009.
[16] T.V. Dunwiddlie, S.A. Masino, The role and regulation of adenosine in the central nervous system, Annu. Rev. Neurosci. 24 (2001) 31–55.
[17] J.W. Dietmer, A.J. Verkhratssky, C. Lohr, Calcium signaling in glial cells, Cell Calcium 24 (1998) 405–416.
[18] M.J. O'Donovan, N. Chub, Population behaviour and self organization in the genesis of spontaneous rhythmic activity by developing spinal networks, Semin. Cell Dev. Biol. 8 (1997) 21–28.
[19] W. Senn, I. Segev, M. Tsodyks, Reading neuronal synchrony with depressing synapses, Neural Comput. 10 (1998) 815–819.
[20] C. Rivera, J. Voipio, J.A. Payne, E. Ruusuvuori, H. Lahtinen, K. Lamsa, U. Pirvola, M. Saarma, K. Kaila, The K^+/Cl^- co-transporter KCC2 renders GABA hyperpolarizing during neonatal maturation, Nature 397 (1999) 251–255.
[21] Z.J. Zhou, A critical role of the strychnine-sensitive glycinergic system in spontaneous retinal waves of the developing rabbit, J. Neurosci. 21 (2001) 5158–5168.
[22] W.T. Wong, K.L. Myhr, E.D. Miller, R.O.L. Wong, Developmental changes in the neurotransmitter regulation of correlated spontaneous retinal activity, J. Neurosci. 20 (2000) 351–360.
[23] Z.J. Zhou, D. Zhao, Coordinated transitions in neurotransmitter systems for the initiation and propagation of spontaneous retinal waves, J. Neurosci. 20 (2000) 6570–6577.
[24] A. Meyer-Franke, M.R. Kaplan, F.W. Pfrieger, B.A. Barres, Characterization of the signalling interactions that promote the survival

and growth of developing retinal ganglion cells in culture, Neuron 15 (1995) 805–819.
[25] R.E. Harris, M.G. Coulombe, M.B. Feller, Dissociated retinal neurons form periodically active synaptic circuits, J. Neurophysiol. 88 (2002) 188–195.
[26] A. Bacci, C. Verderio, M. Pravettoni, Synaptic and intrinsic mechanisms shape synchronous oscillations in hippocampal neurons in culture, Eur. J. Neurosci. 11 (1999) 389–397.
[27] T. Opitz, A.D. de Lima, T. Voigt, Spontaneous development of synchronous oscillatory activity during maturation of cortical networks in vitro, J. Neurophysiol. 88 (2002) 2196–2206.
[28] M.A. Colicos, S.I. Firth, J. Bosze, J. Goldstein, M.B. Feller, Emergence of realistic retinal networks in culture promoted by the superior colliculus, Dev. Neurosci. (2005), in press.
[29] L.C. Katz, C.J. Shatz, Synaptic activity and the construction of cortical circuits, Science 274 (1996) 1133–1138.
[30] E.S. Ruthazer, H.T. Cline, Insights into activity-dependent map formation from the retinotectal system: a middle-of-the brain perspective, J. Neurobiol. 59 (2004) 134–146.
[31] P. Godement, J. Salaun, M. Imbert, Prenatal and postnatal development of retinogeniculate and retinocollicular projections in the mouse, J. Comp. Neurol. 230 (1984) 552–575.
[32] D.K. Simon, D.D.M. O'Leary, Development of topographic order in the mammalian retinocollicular projection, J. Neurosci. 12 (1992) 1212–1232.
[33] L.M. Chalupa, C.J. Snider, Topographical specificity in the retinocollicular projection of the developing ferret: an anterograde tracing study, J. Comp. Neurol. 39 (1998) 35–47.
[34] T. McLaughlin, C.L. Torborg, M. Feller, D.D.M. O'Leary, Retinotopic map refinement requires spontaneous retinal waves during a brief critical period of development, Neuron 40 (2003) 1147–1160.
[35] D.K. Simon, G. Prusky, D.D.M. O'Leary, M. Constantine-Paton, *N*-methyl-D-aspartate receptor antagonists disrupt the formation of a mammalian neural map, Proc. Natl. Acad. Sci. USA 89 (1992) 10593–10597.
[36] A. Bansal, J.H. Singer, B.J. Hwang, M.B. Feller, Mice lacking specific nAChR subunits exhibit dramatically altered spontaneous activity patterns and reveal a limited role for retinal waves in forming ON/OFF circuits in the inner retina, J. Neurosci. 20 (2000) 7672–7681.
[37] C.J. Shatz, M.P. Stryker, Prenatal tetrodotoxin infusion blocks segregation of retinogeniculate afferents, Science 242 (1988) 87–89.
[38] A.A. Penn, P.A. Riquelme, M.B. Feller, C.J. Shatz, Competition in retinogeniculate patterning driven by spontaneous activity, Science 279 (1998) 2108–2112.
[39] P.M. Cook, G. Prusky, A.S. Ramoa, The role of spontaneous retinal activity before eye opening in the maturation of form and function in the retinogeniculate pathway of the ferret, Vision Neurosci. 16 (1999) 491–501.
[40] A.D. Huberman, G.-Y. Wang, L.C. Liets, O.A. Collins, B. Chapman, L.M. Chalupa, Eye-specific retinogeniculate segregation independent of normal neuronal activity, Science 300 (2003) 994–998.
[41] D. Stellwagen, C.J. Shatz, An instructive role for retinal waves in the development of retinogeniculate connectivity, Neuron 33 (2002) 357–367.
[42] F.M. Rossi, T. Pizzorusso, V. Poriatti, L.M. Marubio, L. Maffei, J.-P. Changeux, Requirement of the nicotinic acetylcholine receptor b2 subunit for the anatomical and functional development of the visual system, Proc. Natl. Acad. Sci. USA 98 (2001) 6453–6458.
[43] G. Muir-Robinson, B.J. Hwang, M.B. Feller, Retinogeniculate axons undergo eye-specific segregation in the absence of eye-specific layers, J. Neurosci. 22 (2002) 5259–5264.
[44] S.M. Grubb, F.M. Rossi, J.-P. Changeux, I.D. Thompson, Abnormal functional organization in the dorsal lateral geniculate of mice lacking the b2 subunit of the nicotinic acetylcholine receptor, Neuron 40 (2003) 1161–1172.
[45] E. Sernagor, S.J. Eglen, R.O.L. Wong, Development of retinal ganglion cell structure and function, Prog. Retinal Eye Res. 20 (2001) 139–174.
[46] N. Tian, D.R. Copenhagen, Visual stimulation is required for refinement of ON and OFF pathways in postnatal retina, Neuron 39 (2003) 85–96.
[47] R.C. Stacy, R.O.L. Wong, Developmental relationship between cholinergic amacrine cell processes and ganglion cell dendrites of mouse retina, J. Comp. Neurol. 456 (2003) 154–166.
[48] S.R. Bodnarenko, L.M. Chalupa, Stratification of ON and OFF ganglion cell dendrites depends on glutamate-mediated afferent activity in the developing retina, Nature 364 (1993) 144–146.
[49] B. Chapman, Necessity for afferent activity to maintain eye-specific segregation in ferret lateral geniculate nucleus, Science 287 (2000) 2479–2482.
[50] A.D. Huberman, D. Stellawagen, B. Chapman, Decoupling eye-specific segregation from lamination in the lateral geniculate nucleus, J. Neurosci. 22 (2002) 9419–9429.